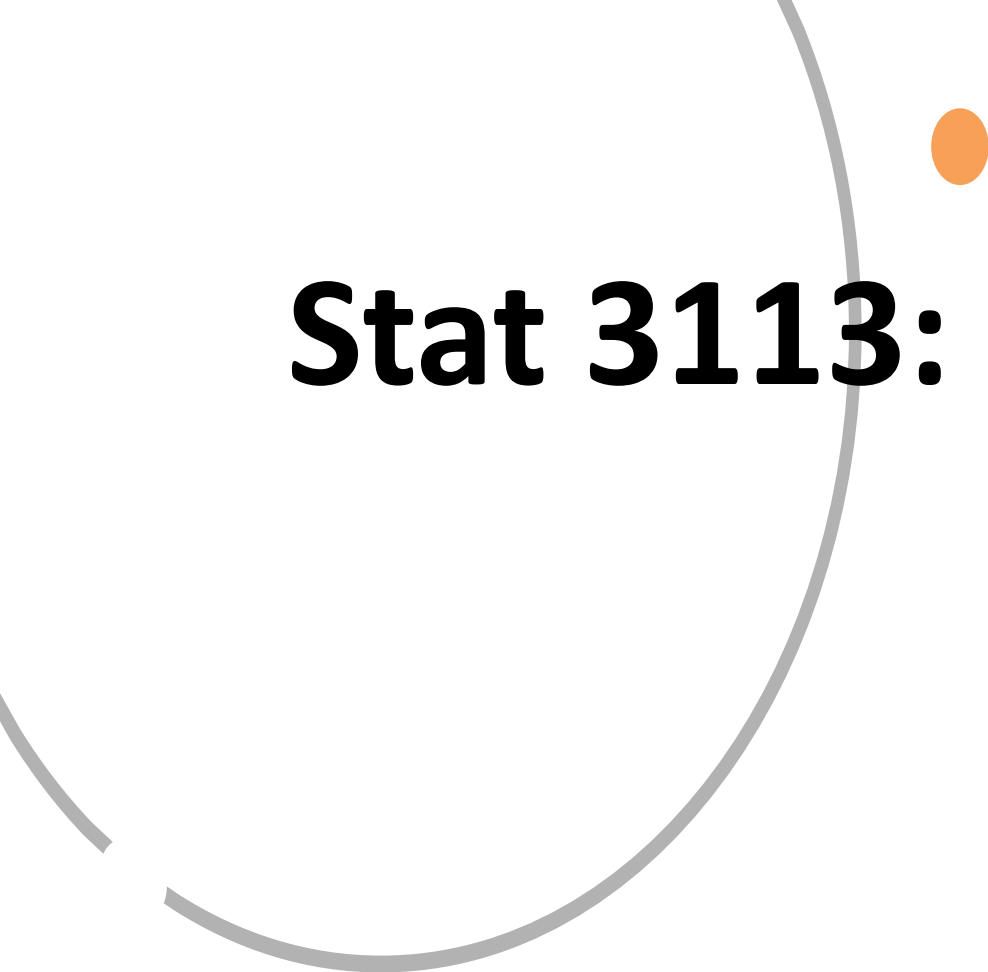
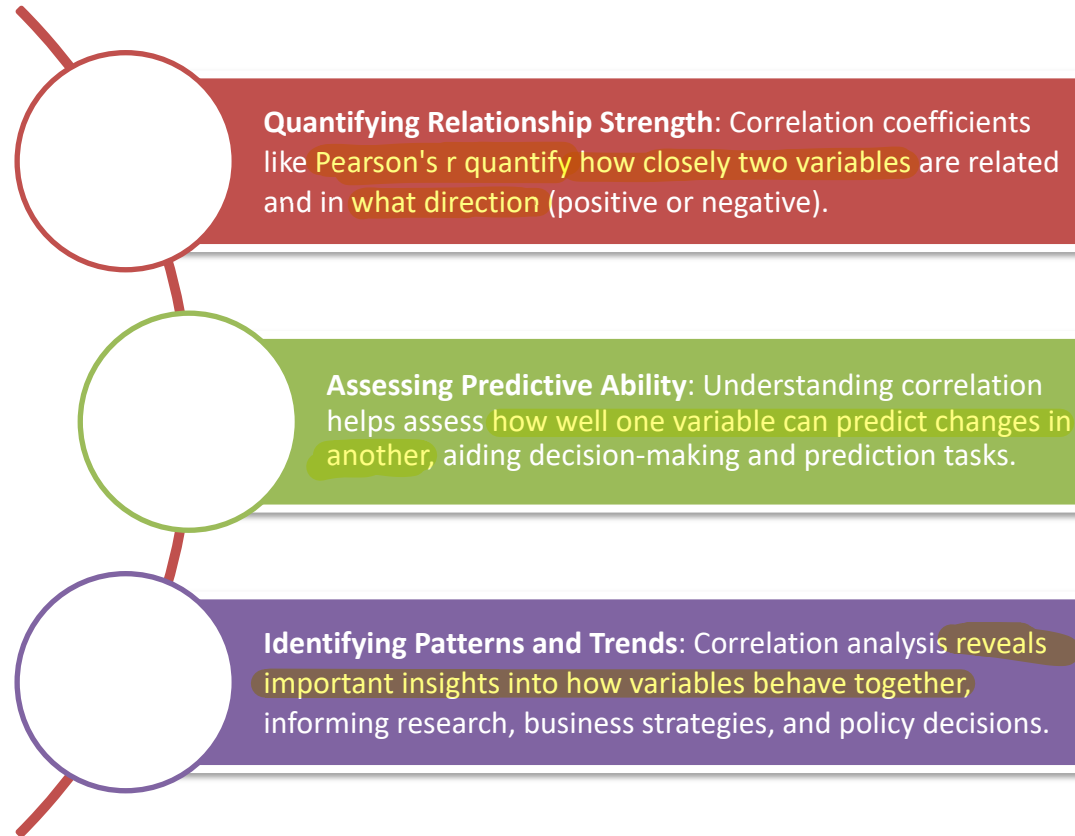




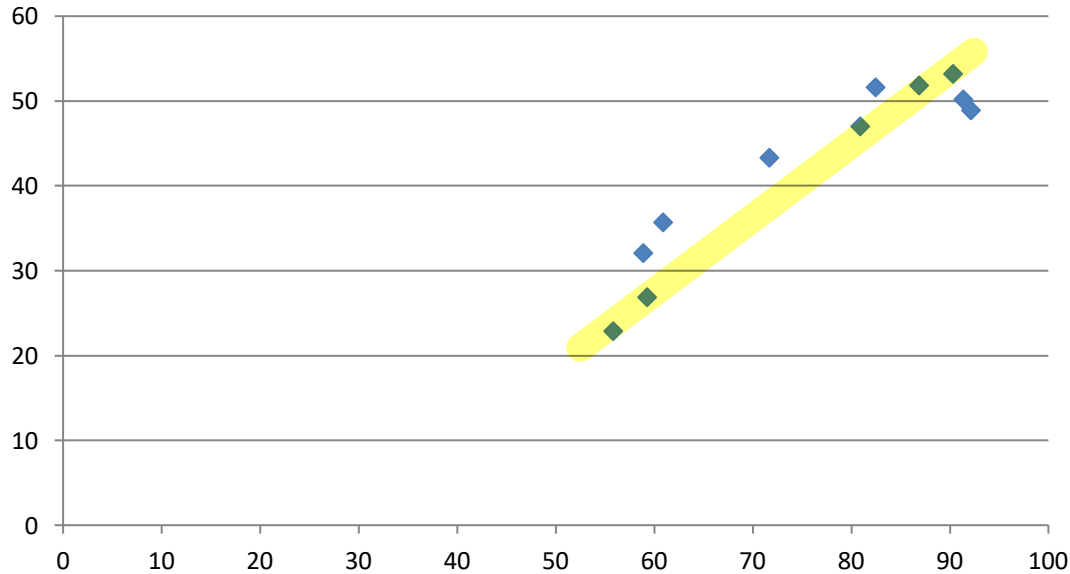
# Stat 3113: Lesson 33

## Correlation

# Motivations



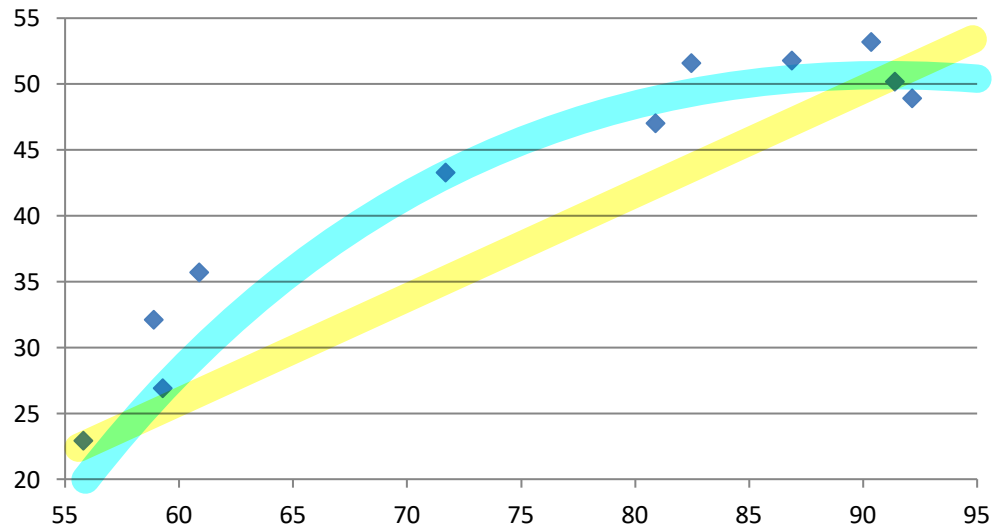
### Scatterplot 1



Which one is more linear?

Which one has a stronger positive association?

### Scatterplot 2



**Correlation (Pearson's correlation)**: Measure of the strength and direction of the linear relationship between two quantitative variables. Denoted  $r$ .

Correlation Facts:

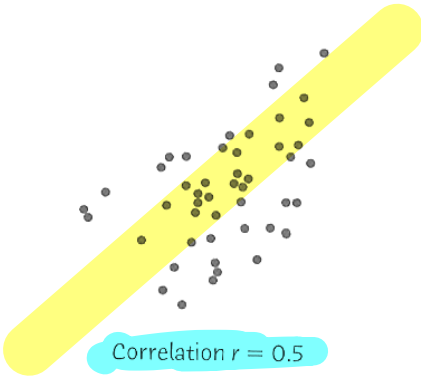
1.  $-1 \leq r \leq 1$
2. Correlation only measures the LINEAR relationship
3. Positive values mean an overall positive association;  
Negative values mean an overall negative association;  
0 means no association (but doesn't mean no relationship)
5. Correlation of 1 or -1, the data form a perfect line. The closer the correlation is to these values, the more linear the relationship is
6. Correlation is impacted by outliers



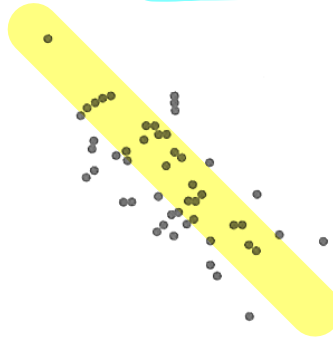
Correlation  $r = 0$



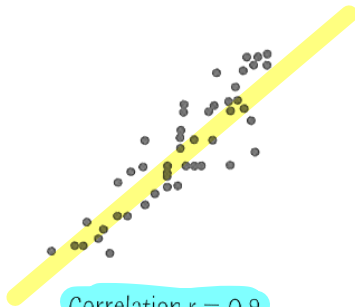
Correlation  $r = -0.3$



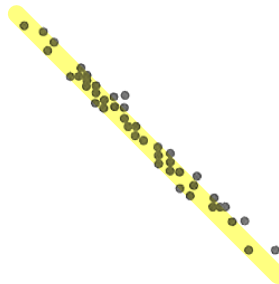
Correlation  $r = 0.5$



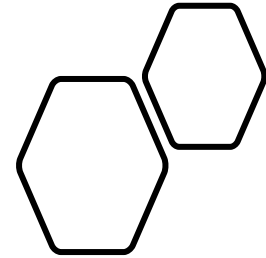
Correlation  $r = -0.7$



Correlation  $r = 0.9$



Correlation  $r = -0.99$

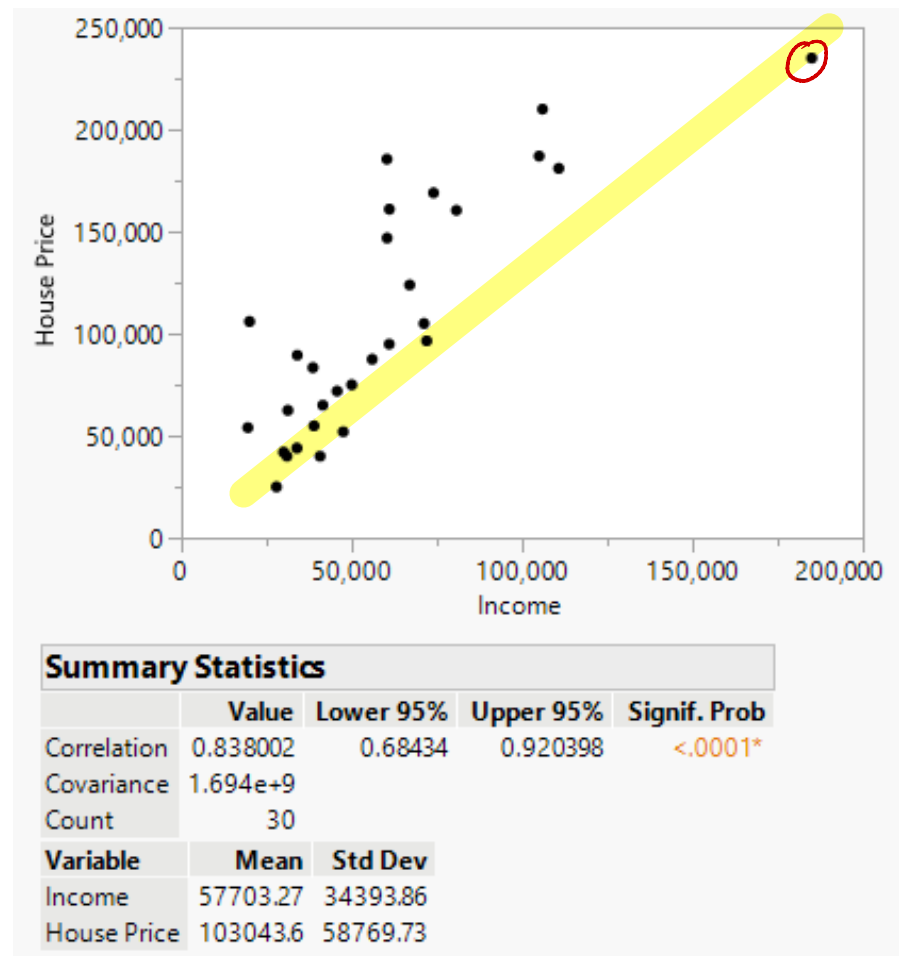


Example:

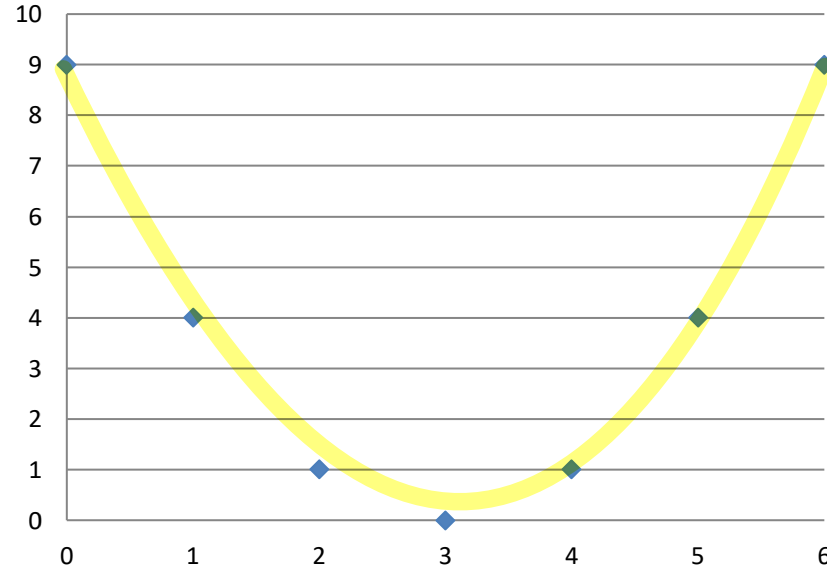
x = gross family income

y = cost of house when purchased

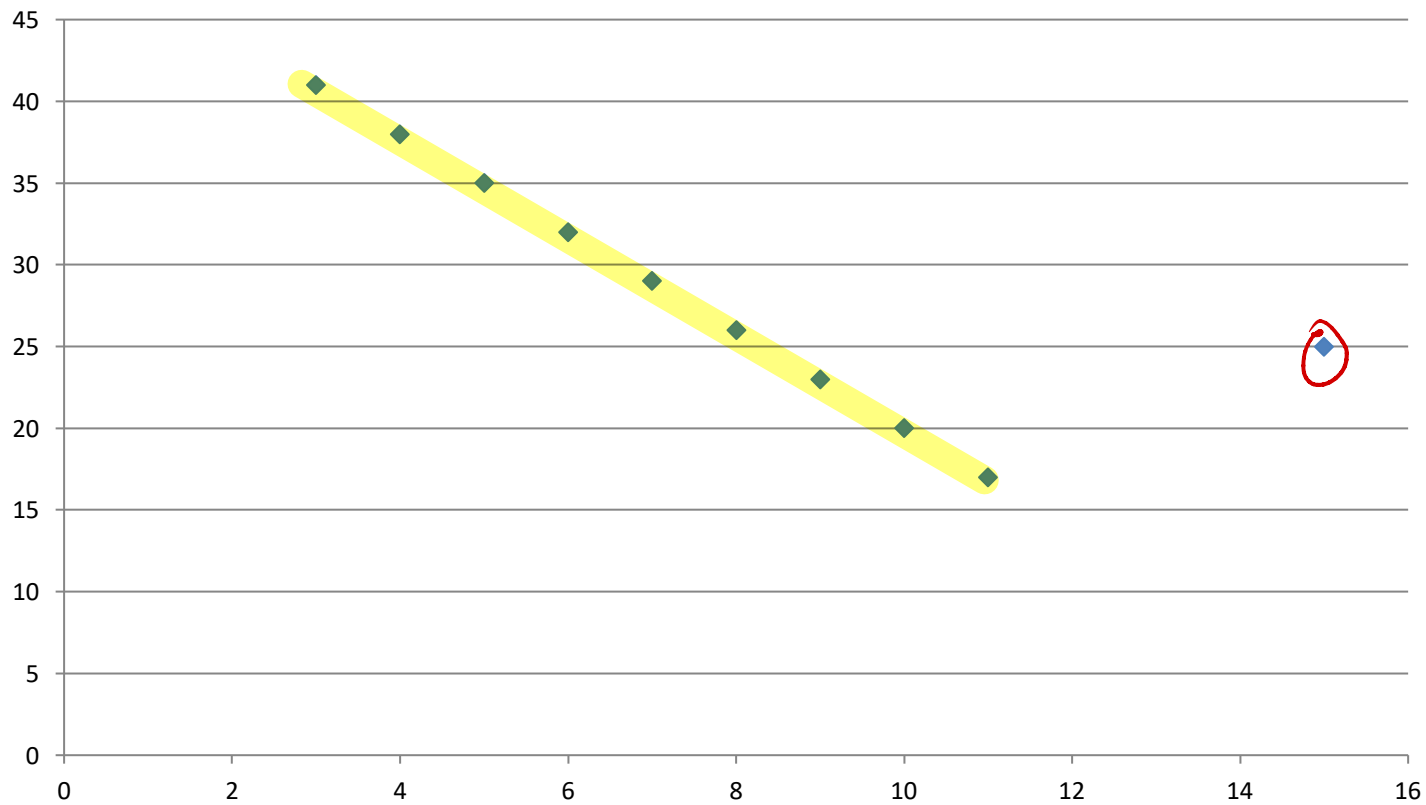
- **OUTLIERS:** *yes, one data point*
- **ANY CLUSTERS:** *No cluster.*
- **FUNCTIONAL FORM:** *Linear*
- **ASSOCIATION:** *Positive*



# Examples:



**$r = 0$  -> Pearson's correlation does not measure anything but linearity**



$$r = -0.81774$$

**YOU MUST USE SCATTERPLOT AND CORRELATION VALUE TOGETHER!**



# Reading

- CHAPTER 2 – SECTION 2.1

# Next Topic



Chapter 2 – Sections 2.2, 2.3

Chapter 8 – Sections 8.1, 8.2



Regression